Challenges and Limitations of TVAE Tabular Synthetic Data Generator

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Abstract. Synthetic data is commonly used today in machine learning. From the traditional statistical approach of imputing missing information, the data generation techniques have now leaned towards machine learning and artificially intelligent generative techniques. Generative Adversarial Networks (GAN) and Variational Autoencoder (VAE) are the techniques employed. Tabular data is most common in industrial research. TVAE generates synthetic data which is tabular and available as a part of synthetic data vault (SDV) introduced by Neha Patki et al. Hence, we pick TVAE, perform an in-depth analysis on existing literature and experimentally highlight the strengths and limitations of TVAE, conducted using different types of tabular data. TVAE as such, completely ignores the minority samples in the dataset. Hence, we modify the data generation algorithm by introducing another set of encoder and decoder to oversample only those categorical sub classes which is small in representation and provide that information to the latent space. Our experiments reveal that, despite the several techniques proposed in various literature regarding sampling and balancing the data using VAE, TVAE struggles in producing the categorical values for a subcategory and the associated target class within a feature when the sub-categorical value is a minority. Based on our experiments and analysis, we see a need for further study on modeling the TVAE to produce a well-balanced dataset.

Keywords: Table Variational Autoencoder · Synthetic Data Vault · Generative Adversarial Networks · SMOTE · ADAM

1 Introduction

The concept of synthetic data was first proposed by Donald B Rubin [1]. However, as machine learning and artificial intelligence gained popularity, the techniques of generation shifted towards artificial intelligence. The inception of Generative Adversarial Networks (GAN) [2] led to numerous works around synthetic data generation using GAN [3, 4]. In similar vein, Neha Patki et al. [5] presented Synthetic Data Vault (SDV) to build synthetic data generatively. Table Variational Auto Encoder (TVAE) is one of the libraries available within SDV package. It is based on Variational Auto Encoder (VAE) which was originally proposed by Diederik P. Kingma and Max Welling [6].

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Figure 1 shows the architecture of TVAE which employs two independent modules known as Encoder and Decoder. Encoder extracts the mean, variance and standard deviation from the real data provided during the training and updates the information in the latent space. Encoder uses functions such as softmax, cross entropy and Kullback-Leibler Divergence (KLD) as a measure to improve prediction and the information gets updated in the latent space. The decoder uses the noise which is the summation of the matrix of zeros (mean) and ones (standard deviation) of defined size, extracts the statistical information from the latent space to reconstruct the data that matches the input data.

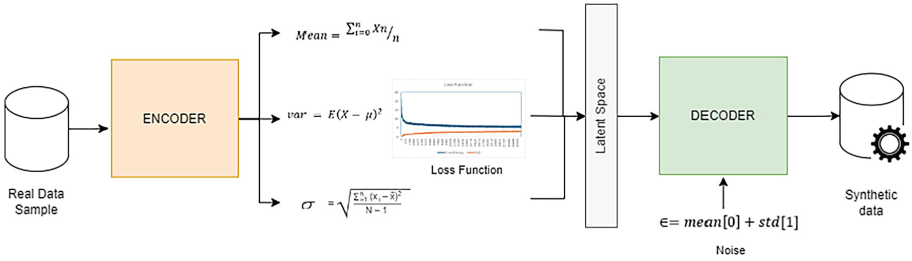


Fig. 1. TVAE Architecture

Generally GANs are considered to produce superior images and are better suited for image generation but they pose several challenges with high dimensional parameters [7]. Conversely, TVAE is much simpler, easier to interpret [8]. TVAE also displays better efficiency. However, the limitation observed, as highlighted in their work by A. Kiran and S. Saravanakumar [9] is, TVAE encounters difficulties when creating synthetic data in situations where data is not balanced. TVAE completely ignores the representation of any subcategory of a categorical column in the training dataset. This presents a significant challenge to the resultant datasets as most of the subcategories get eliminated. This happens because the models make assumptions and over simplify the posterior distribution [10]. In the real world, the healthcare data and financial domain data in specific will have significant data imbalance. Hence it becomes important to explore how the TVAE synesthetic data generator can be improved.

Therefore, the objective of our work is

- to analyze the prior contributions related to VAE
- enhance the TVAE algorithm, which is achieved by picking specific sub class in the categorical column that are represented in small number
- oversample the representation and put them into the latent space so that decoder has better sample representation to reproduce from, enabling it to produce synthetic data, even in scenarios where there exists a substantial disparity between the larger and smaller representation of categories within the feature of the original dataset.

The sections to follow in this paper will delineate the relevant literature, methodology, experimental procedures encompassing the setup and dataset employed, the obtained results, and finally, the conclusion.

2 Related Work

The short coming of VAE was first highlighted by Burda et al. [10] where they analyze that the VAE just assumes and simplifies the distribution rather than considering the entire model. They suggest a weighted model by lowering the weightage of the decoder or applying more constraints to recognize the distribution rather than making assumptions. To improve the learning in the VAE, S. Zhao et al., [11] suggest using maximum mean discrepancy (MMD) instead of evident lower bound (ELBO) that uses KL divergence. They concluded with the experiments on MNIST data that MMD is able to learn more information from the latest space. To address the specific issues related to imbalance in the data set, Y. Zhao et al., [12] utilized the information from the entire dataset to oversample the minority classes. They leveraged VAE algorithm to perform oversampling and referred to their approach as Conditional VAE with self-transferred algorithm (CVAE-SeTred), as the knowledge and instructions were transferred from the majority to the minority class. In their approach the minority and majority sub classes were split, two sperate encoders were trained independently to generate the latent data which is then used by the decoder for recreation. All these techniques focus on image classification, not on tabular data. Fajardo et al., [13] introduce a two 2 stage latent structure, one to address only the independent variables of the dataset "x" and the other one which takes the distribution of both independent "x" and dependent variable or the target class "y". Experiments are focused on the accuracy of the machine learning algorithm using oversampled data. It does not conclude if all the other subcategories in the data set was reproduced as desired. Method introduced by Chunakai Zhang et al. [14] also discuss oversampling only the target class**.** Tab-VAE presented by Tazwar et al. [8] proposes to use "softmax" instead of gaussian distribution on categorical or discrete values which TVAE has already adopted. With the proposed approach, we intend to achieve a good representation of the less observed value in every column of the resultant tabular dataset. To achieve this, we introduce a new layer of encoder and decoder, make modifications to the TVAE class and analyze results.

3 Methodology

As shown in Fig. 2, we generate synthetic data using TVAE in three iterations. In the first iteration, synthetic data is generated as-is without performing any modification to the TVAE. In the second iteration, we introduce the sampling logic, which is explained in Sect. 3, Fig. 3. In the third iteration, the real data is balanced using SMOTE [15] and then used in generation of the synthetic data with TVAE. The resultant data sets generated in the three iterations are compared to validate how the proposed modification to the TVAE performs and if it can achieve the intended objective.

4 Experiments

Our goal is to pick each subcategory from columns with discrete data and check if the samples within the subcategory have enough representative samples. If a sub class is found to have lesser representation, then oversample it, just enough to have a fair

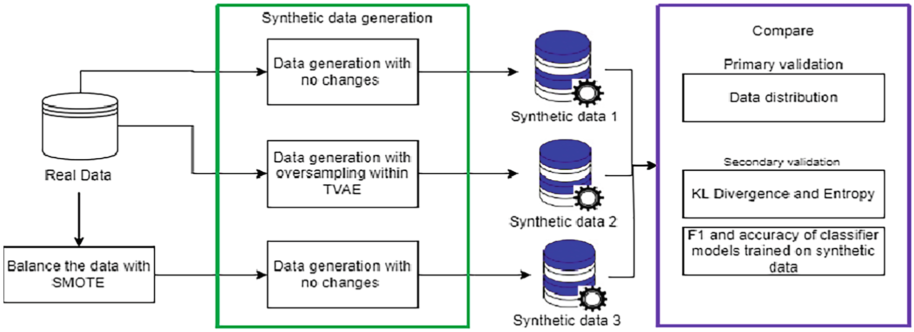


Fig. 2. Methodology used in experiments

representation. This methodology is repeated for each column. As can be seen in Fig. 3, the TVAE is modified to use two encoders and two decoders. The first pair of encoder and decoder are used to sample the subcategories with less representation. Once the samples are created, it is concatenated with the original dataset and then the second encoder is trained with the consolidated dataset.

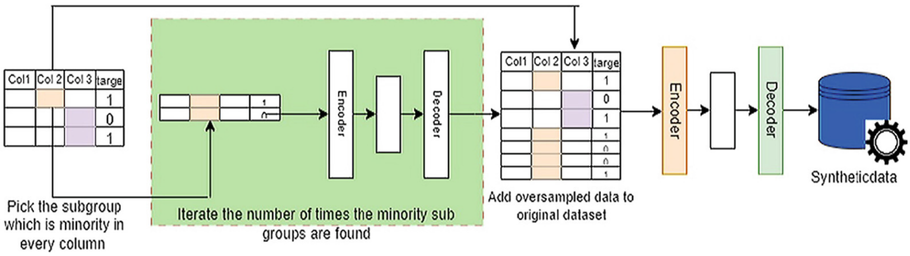


Fig. 3. An oversampling methodology to have a fair representation of minority subgroups in the resultant dataset generated using the TVAE synthetic data generator

4.1 Algorithm

The algorithm is as follows

Let

N be the number of rows in the dataset.
X be the input sample dataset
$x_{majority}$ be the dataset with all the tuples relating to majority target class.
$x_{minority}$ is the intermediate dataset with tuples pertaining to only minority data
$n_{minority}$ represent the number of counts of the minority class.
$columns_{Unique}$ Unique subcategories within the column

Sampling Algorithm

Define sampling_weightage (decimal value between 0 and 1)
for *all columns in dataframe*
get all $columns_{types}$
get $columns_{Unique}$
if $columns_{Unique}$ *is BETWEEN 2 and 6 (If the subcategory is only one, its not categorical, hence it is discarded)*
get the count of discrete values within $columns_{Unique}$
$n_{minority}$ = *count of values in* $n_{minority}$ / *length of dataframe*
if $n_{minority}$ < *1/total number of categories*
get all minor class values which is $x_{minority}$
*sampling_size = (N – (2** $x_{minority}$*)) * sampling_weightage*
$x_{oversampled}$ = *Fit the new layer of encoder and decoder with* $x_{minority}$ *with the desired sampling_size*
$x_{train} = x_{oversampled} \cup x_{majority}$
end if
end if
end for
Fit TVAE with x_{train}

We override the fit function of TVAE by introducing our algorithm. The columns with categorical values are picked and the ratio of count of records in every category is checked in relation with the total count of records in the dataset. If the ratio is less than the fraction of total number of categories, then those samples are used and fed into another encoder/decoder pair for oversampling.

Second parameter considered is the amount of oversampling. An established approach is, the maximum size for minority sampling is equivalent to the size of the majority dataset, ensuring an equal ratio between the two classes. However, this approach may result in an overly balanced dataset, which is not the desired outcome. The goal is to generate just enough samples while still maintaining the imbalance to resemble the real dataset, rather than achieving a perfectly balanced dataset. To achieve this, a sampling weightage factor is utilized that determines the desired number of samples of the minority sub class in the resulting dataset. This weightage is specified during the generation process. The integer and float values are not considered except for target class.

4.2 Data

Table 1 presents the sample data that was used in our experiments. Each of the dataset is judiciously chosen based on their target class distribution, the data types and the combination of discrete, continuous and categorical column to test how our algorithm would perform as compared to the non-modified TVAE synthetic data generator.

Table 1. List of datasets with their specifications used as samples to generate synthetic data using TVAE.

Dataset	Data types	Rows	Features (Columns)	Discrete columns	Target class	Target %
Stroke Dataset	categorical, continuous, discrete	43401	12	8	Biopsy	1.80
Cervical Cancer	discrete and categorical	858	36	31	Target	6.41
Credit Card Approval Prediction	discrete and categorical	25129	21	15	Status	0.49
Credit Card Farud	continuous	284808	31	1	Class	0.17
Titanic	continuous, discrete	891	12	5	Survived	38.38
HR Analysis	categorical, discrete	21287	14	7	Target	24.93

Verifying if the TVAE is able to generate all the right samples of the sub categorical classes in the dataset is the primary objective whereas the secondary validation is to check the generic and specific utility [16] of the generated data. Generic utility is validated using a two-step approach. 1) Verifying the presence of the sub categorical values with respect to its target class and, 2) Compare the Correlation map, Entropy and KL divergence in relation to the real data. The specific utility is validated by comparing the accuracy of logistic regression algorithm trained with the real data and synthetic data generated using the TVAE.

5 Results

The evaluation is carried out by comparing the generic utility, which is the statistical resemblance of the synthetic data set with the original data and then the specific utility which is comparing the accuracy of ML algorithm trained using the synthetic data. This section tabulates the results in chronological order. First, we compare the generic utility of the synthetic dataset generated using the three different iterations as explained in the methodology section. Table 2 tabulates the column with the categorical data that was used for assessment as a sample. Cervical cancer and credit card fraud datasets did not have any categorical columns to compare.

Figure 4 illustrates the comparison of the categorical columns in the real dataset (a), Synthetic data generated using TVAE after balancing the real dataset using SMOTE (b) and the data generated using TVAE generator using the custom sampling technique.

Cerebral stroke and HR Analysis datasets are shown as illustrations. The plot provides a relationship between categories of work type against having stroke. The bar depicts the

Table 2. Comparison of column with categorical data in the real dataset and the categorical class that was never produced in the synthetic data set generated using the TVAE generator.

Data set	Column considered for validation	Categories in original dataset	Categories lost in generation
Stroke Dataset	Work_type	Children, Private, Never_Worked, Self_Employed, Govt_Job	Govt_Job
Credit Card Approval Prediction	Income_type	Working, Commercial associate, State Servant, Student, Pensioner	Pensioner
Titanic	Embarked	S, C, Q	
HR Analysis	Company_type	Pvt Ltd, Funded Startup, Early-Stage Startup, Other, Public Sector, NGO	Public Sector, NGO

number of occurrences of the respective categories in the dataset. Table 2 provides the tabulation of all the datasets and the corresponding columns chosen in the experiment.

Table 3 provides the comparison of the Euclidean distance of the categorical columns picked as sample to validate the datasets. Euclidean distance calculates the distance between two points. The distance of the chosen column value is compared between real data and synthetic data.

After the comparison of individual subcategories in the categorical columns, we also compare the utility of the generated data using KL divergence which measures the distance between probability distribution of the real data and the synthetic data and also the entropy, which is the error introduced in the generation. Table 4 provides the values observed for the same.

The comparison of correlation matrix for real and the synthetic data generated for cerebral stroke using TVAE with the customized oversampling algorithm is shown in Fig. 5a and 5b and Fig. 5c provides the comparison of roc_auc curve. Table 5 provides the roc_auc scores for all the datasets used in the experiment.

Finally, the synthetic data is generated using GAN based generator from the SDV package to compare how the algorithm introduced for TVAE perform against GAN based generator. Figure 6 shows the data distribution for categorial columns for Real data, Synthetic data generated using TVAE with enhancement and then the data generated using DPCTGAN.

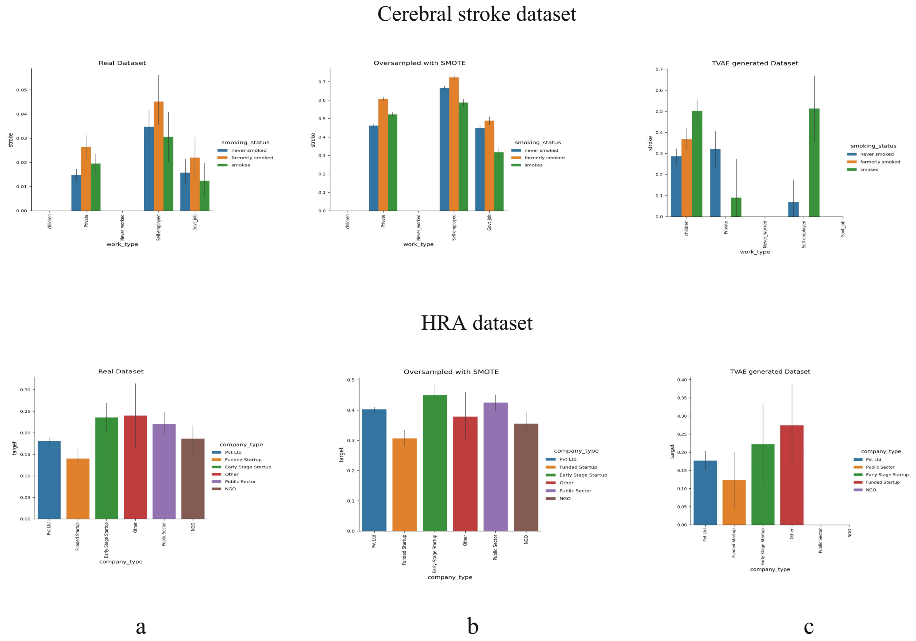


Fig. 4. Comparison of the Cerebral Stroke real data, synthetic data generated using TVAE after balancing the input data with SMOTE and TVAE generated synthetic data with the new oversampling technique that is introduced in this paper. Comparison shows different categorial classes of work_type column and their respective distribution against the target class for cerebral stroke dataset and the company_type against the target for HRA data.

Table 3. Euclidean distance comparison of specific categorial column against datasets generated using the three different variations

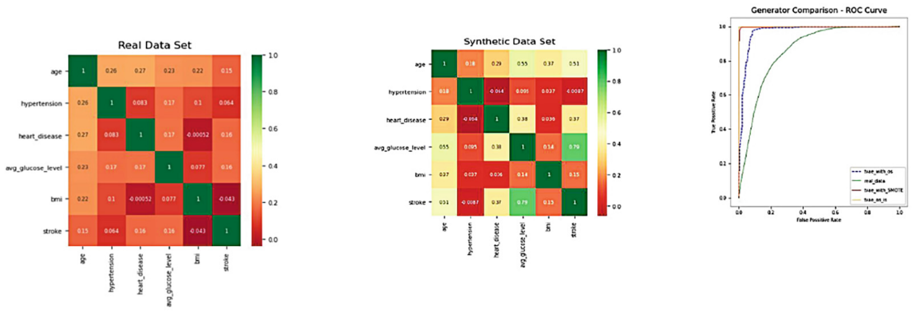
Datasets	Column	TVAE AS-S	TVAE with Sampling	SMOTE-TVAE
Stroke Dataset	work_type	64.490309	71.805292	87.441409
Credit Card Approval Prediction	Income_Type	95.451558	98.325988	119.230868
Titanic	Embarked	22.135944	26.720778	0
HR Analysis	Company_Type	84.017855	84.397867	83.234608

6 Discussion

Our experiments and comparison of generic and specific utilities of the real dataset and the synthetic dataset generated using three different techniques with TVAE generator reveal that the TVAE generator as-is with no modification fails to generate the subcategories and target class that are less in representation, which is evident from Table 4, where

Table 4. Tabulation of *Kullback-Leibler Divergence* and *entropy* for the datasets generated using three different techniques.

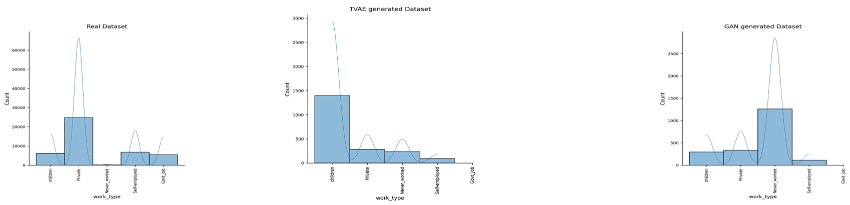
	TVAE AS- IS		TVAE with Sampling		SMOTE-TVAE	
Data set	KLD	Entropy	KLD	Entropy	KLD	Entropy
Stroke Dataset (age)	68152.777	0.7902	32121.705	0.37142	68123.342	0.277081
Cervical Cancer (age)	2200.5719	0.0951	2411.3052	0.10457	2056.7322	0.08391
Credit Card Approval Prediction (Good Debt)	26319.333	0.7471	27297.423	0.53575	37899.916	0.522672
Credit Card Farud	No categorical columns to compare					
Titanic (age)	5606.1166	0.2116	7066.408	0.26686	inf	inf
HR Analysis (training hrs)	81495.247	0.5017	106904.371	0.66881	98826.919	0.431196



a b c

Fig. 5. Comparison of correlation map for cerebral stroke real dataset (a) and synthetic data (b) Fig. 5c is the ROC Curve of Logistic classifier algorithm trained on real data, tvae generator as-is, the tvae generator with custom sampling logic and finally data generated using combination of smote and tvae

the target class were never generated. The data generated using the TVAE with the algorithm provided under Sect. 4 which is a variation of technique provided by [12] Y. Zhao et al. Looks good with all the metrics such as Euclidean distance, KLD, Entropy, Correlation and the specific utility using ML algorithms accuracy scores. However, when the data set is analyzed more closely by comparing individual categories and subcategories within the dataset, we observe that the minority representation is not fully reproduced. Table 2 tabulates the categories that is completely discarded during the data generation. This is also evident from Fig. 4 showing the comparison for cerebral stroke and



(a) (b) (c)

Fig. 6. Comparison of the distribution of categories found in work_type column in Real Data (a) and synthetic data generated using the enhanced algorithm of TVAE (b) and DPCTGAN (c)

Table 5. Comparison of ROC_AUC scores of the real data with synthetic data generated using three different techniques

Datasets	Real Data	TVAE AS-IS	TVAE with Sampling	SMOTE-TVAE
Stroke Dataset	0.8576996	DNG	0.9679287	0.9956866
Cervical Cancer	0.9788728	DNG	0.9801090	0.9460171
Credit Card Approval Prediction	0.9978564	DNG	0.9934098	0.9423336
Credit Card Farud	0.9887823	DNG	0.9969780	0.9571368
Titanic	0.8493087	0.8652266	0.9205976	0.8318623
HR Analysis	0.7496894	0.7116442	0.7510220	0.6450947

HRA datasets. Table 3 provides the information for other datasets used in experiements. The processing also takes more time as there is an additional oversampling step that is introduced. The is considerable when the number of categorical columns increase.

During the process of data generation, modifications were made to the optimizers to assess their impact on the resulting data. Among the various optimizers utilized, the ADAM produced the best results. The *betas*, *weight_decay, amsgrad* and *epsilon* values were not altered. Except the learning rate no other parameter had a major impact on the resulting outcomes. Similarly, alternative optimizers such as NADAM, SGD, Adagrad and RMSProps were also tried. There were no significant changes in the results.

The approach used is to pre-process or balance the sample data using SMOTE and then using TVAE generator to generate the synthetic data produced satisfactory results as seen in Fig. 4, Table 3 and Table 4. Finally, from Fig. 6, we observe that the enhanced algorithm of TVAE provides a comparable result with the data generated using GAN based generator.

7 Future Scope of Work

The objective to produce enough samples of the subcategories with minority representation is achieved with the enhanced algorithm. However, there are still inconsistencies in the generation that must be explored further, specifically focusing on the information in the latent space that could help the decoder in better reproducibility of real data. Second area is to improve the efficiency, as the number of categories increase, the resource utilization increases. Third area of focus is to understand impact of oversampling on overfitting, as observed from the scores tabulated in Table 5, the models tend to overfit when the minority samples are extremely minuscule as in the case of credit card datasets.

8 Conclusion

TVAE synthetic data generator produces synthetic datasets which are like real datasets when the dataset is well balanced. This is evident from the experiments that we conducted both using balanced dataset and also, when the balance was achieved using SMOTE prior to feeding the samples to TVAE for data generation. However, when there is an imbalance, TVAE still struggles to produce datasets that are acceptable. Introducing the new algorithm to oversample the minority categories does produce favorable results as it is tabulated in the results section where most of the generic utility and specific utilities provide comparable results with that of real data. However, a detailed analysis of the underlying structure of the dataset still shows that the categories that has lesser representation are not reproduced. Therefore, we recommend further investigation on TVAE, specifically the latent space parameters so that the decoder can learn better even from the minute sample to produce datasets that match the real data samples. For users of synthetic data who want to use TVAE for its ease and efficiency, balancing the input sample before the generation is recommended.

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